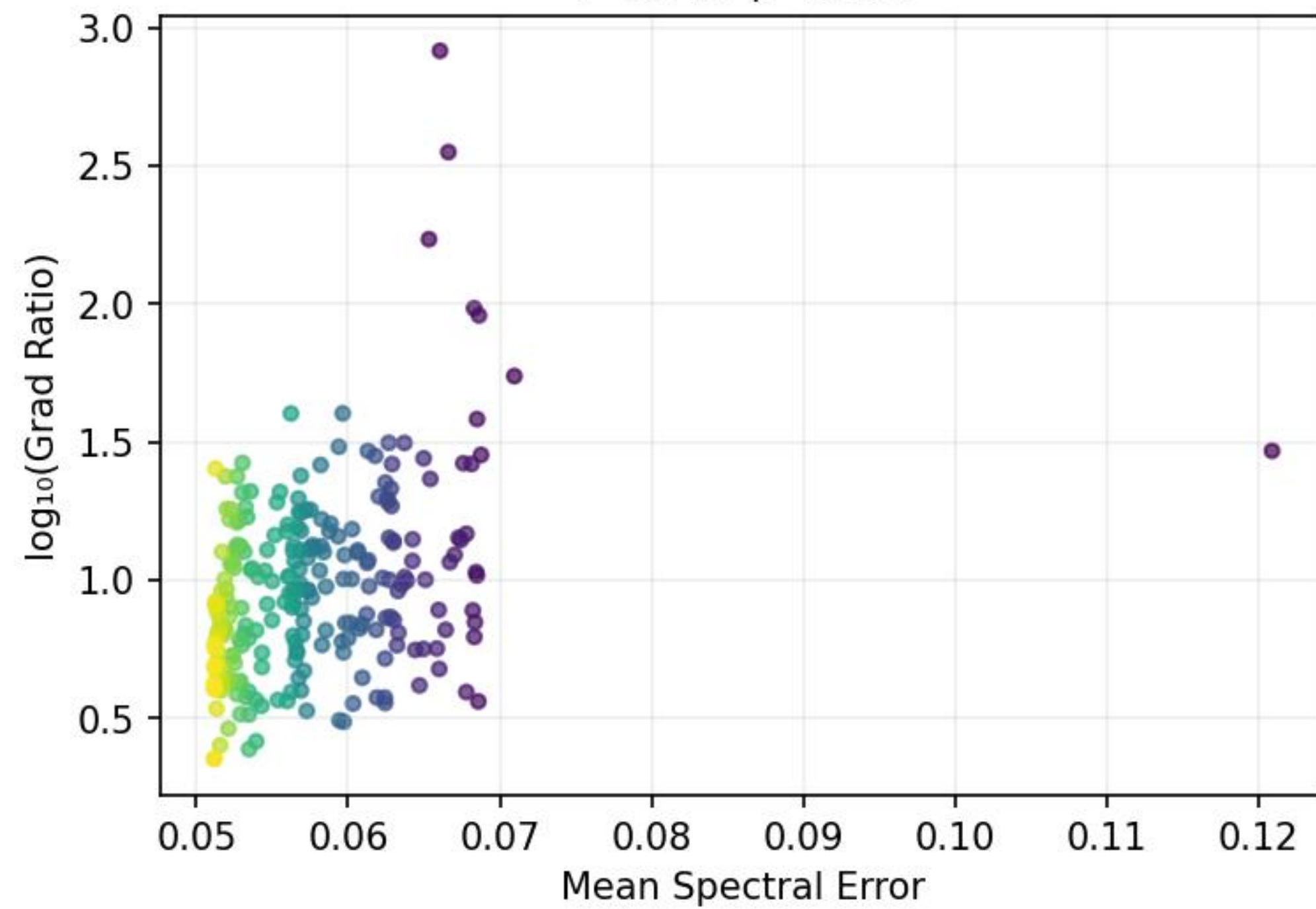
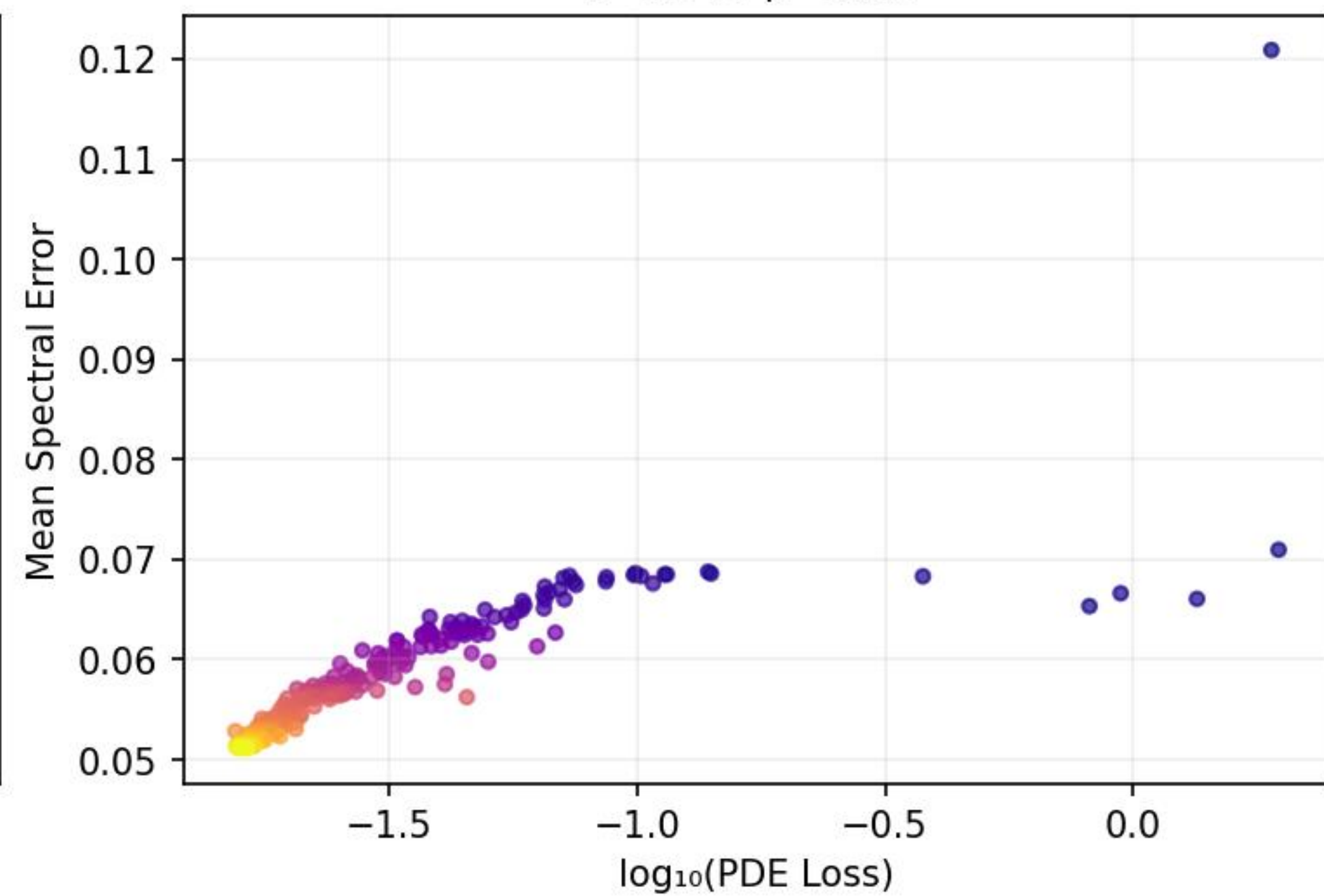


Correlation & Causal Link Analysis ($\beta=30$)

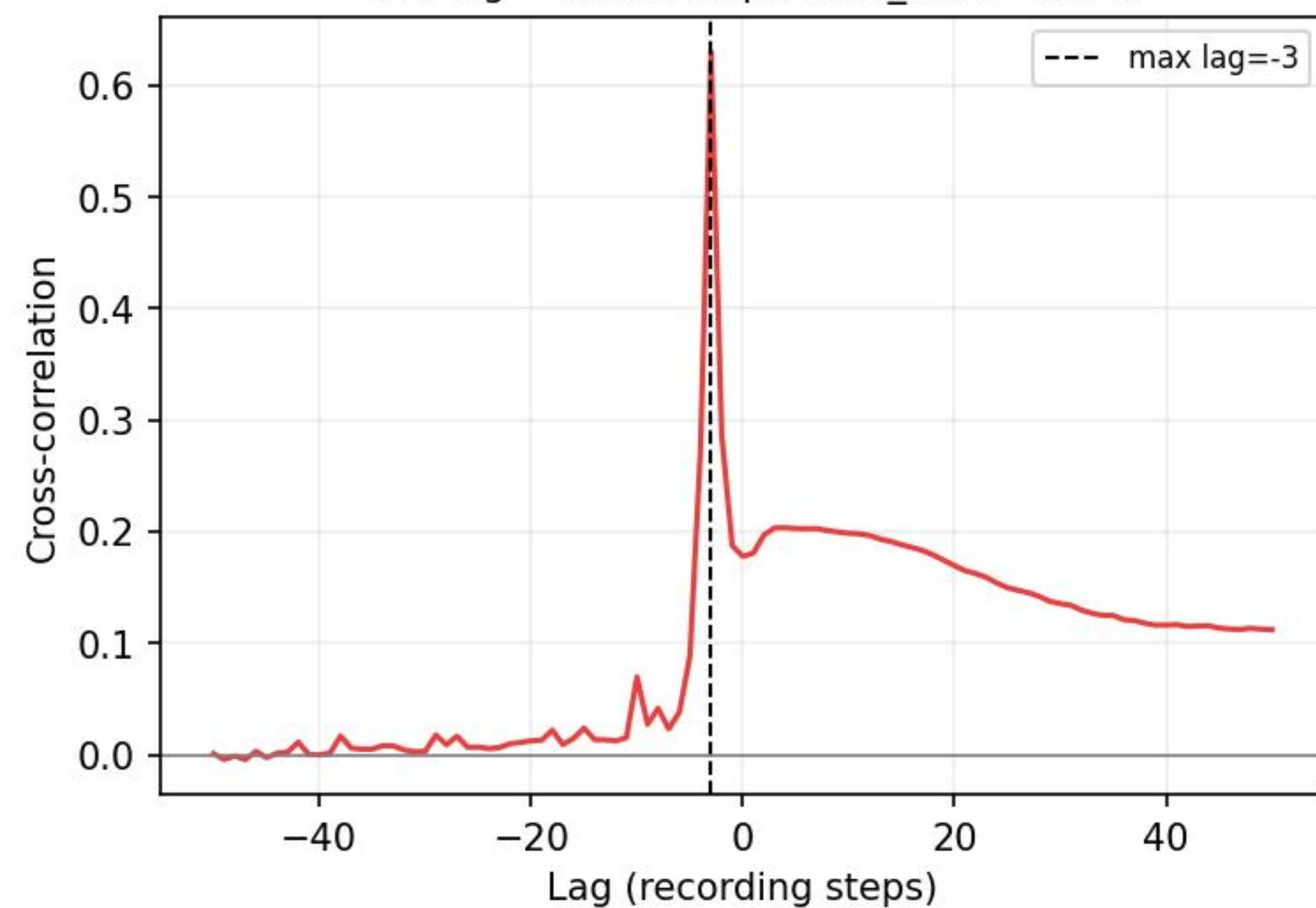
Spectral vs Gradient Ratio
 $r=0.348$ $p=0.000$



PDE Loss vs Spectral Error
 $r=0.842$ $p=0.000$



Spectral $\leftrightarrow$ Gradient Cross-correlation
Peak lag=-3 $\times$ 200 steps max_xcorr=0.630



Failure Mode Correlation Matrix

